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Final Design Report

Team 485: Autonomous Weeding Robot

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1 Executive Summary

Manual weeding is tedious and labor intensive, and chemical methods used for weed removal are harmful to the environment and consumers. The purpose of this report is to provide background on the Autonomous Weeding Robot project and to outline current progress and possible future work. The goal of this project is to minimize physical labor, cost, and harmful herbicide use that effects the environment. The purpose of this project is to design and build an autonomous weeding robot that is easily usable in home gardens and small farms. The robot will implement a Go-To-Goal control system which allows it to identify a "goal," in this case a dandelion or a red target for testing, and go to that goal. Computer vision and machine learning will be used to train the robot to learn to identify dandelions in a garden or farm. The robot will also have a form of waypoint path following with the idea to be used for repeated travel through crop rows. There will be a multi-tool system that will allow the user to easily change tools. The tools include an auger and a tine weeder, and will be expanded upon in future iterations of the project. The client scope involves creating an autonomous robot with a multi-tool system. The Go-To-Goal and waypoint path following implementation and trained model meets autonomous needs as the robot will independently identify a dandelion weed and move towards it. The multi-tool covers a more diverse array of weed removal, including various ages of weeds and garden conditions which satisfied this client requirement. A general recommendation to continue the project is to further advance it's current design. This includes heightened sensitivity in dandelion identification, increased accuracy in autonomous movement, and designing additional tools to interface with the multi-tool attachment system.

2 Introduction

The purpose of this report is to provide background on the Autonomous Weeding Robot project and to outline current progress and possible future work. For this project in particular, this report outlines the primary aspects of the project, being autonomy and controls, machine vision and machine learning, mechanical design, and business analysis. This project seeks to fill a gap in agricultural robotics for small farm level consumers, and to lower the barrier related to physical labor for hobby level use. Current solutions exist for large-scale monoculture farms, but few solutions are in the market for more diverse and smaller farms.

This report covers the problem being solved, the design description across the different aspects of the project, various design evaluations and technical analysis, an evaluation of the project's engineering specifications, and a budget analysis, and recommendations to implement for the future of the project. This report also includes a reflection section, including a retrospective on the project and team, a review of open issues and project satisfactions, and lessons learned through this project.

3 Problems Definition

Weeds reduce crop productivity by competing with them for nutrients, water, and sunlight. Weeds can also make a garden much less visually appealing. Weed management is a persistent challenge across all scales of agriculture, from industrial farms to personal

gardens. Current solutions to this problem are either cost-intensive, require a high capital investment in commercial equipment, use harmful chemicals, or are labor-intensive and require regular manual effort to maintain weed-free growing conditions. These trade-offs reduce accessibility for many potential users. One of the project clients, Dr. Hamid Ossareh, had firsthand experience with this problem in his own garden. Together with the other client, Dr. Luis Duffaut, he realized that their expertise in autonomy could fill a gap in the market while also eliminating Dr. Ossareh's least favorite part of gardening. While commercial robotic weeding systems currently exist, few affordable solutions serve diverse, smaller-scale agricultural environments. The clients seek to develop a low-cost robotic platform that can reduce the labor burden of weed management while remaining accessible to users with limited budgets.

Additional stakeholders other than the project clients include small farmers, personal gardeners, environmentalists, and the engineers working on the project. Small farmers benefit from the reduced labor costs and improved crop yields through more consistent and versatile weed management. Personal gardeners, particularly those with reduced mobility, would gain access to a more efficient solution that minimizes the physical demands of gardening. Environmentalists also have an interest in effective mechanical weed removal as an alternative to chemical herbicides, which can have negative effects on all parts of the environment. The engineering team has a stake in developing a technically sound and professionally executed capstone project to succeed in the class, as well as to have a useful learning experience.

The final scope of this project is to develop an autonomous robotic platform capable of identifying and mechanically removing weeds in lab settings. The robot is intended for outdoor use in typical garden and small farm conditions, and should be capable of operating on farm field terrain and varying soil types. The system will only operate in well-lit situations to support camera-based weed detection. The design will focus on common weed species, and will be trained and set up to be optimized for dandelions and crabgrass. The scope of this year's iteration of the project focuses on the development of a robust mechanical multi-tool system, improving the quality of electrical systems, and building the foundation of autonomous behavior.

The customer requirements for this project, organized by priority, are as follows:

1. **Autonomous Operation:** The robot must autonomously navigate a defined area using path following, while using machine vision to identify weeds. The robot must autonomously navigate to identified test targets.
2. **Accurate Weed Identification:** The system must reliably distinguish target weeds from crops and other objects to prevent damage to desired plants.
3. **Effective Mechanical Removal:** The robot must successfully remove or disrupt weeds at multiple growth stages, from seedlings to mature plants with established root systems.
4. **Support for Multiple Tools:** The mechanical system must have a multi-tool system to address different weed types and growth stages.
5. **Robust Mechanical Design:** The platform must withstand outdoor environmental conditions and the mechanical stresses associated with soil penetration and weed removal.

6. **Low Cost:** The final system must be cost-competitive with existing weeding solutions to ensure accessibility for the target market.

4 Design Description

To solve the problems outlined above, an autonomous weeding robot is proposed. This robot aims to integrate machine vision and autonomous control to target weeds and mechanically remove or disrupt them.

4.1 Autonomous Control

The robot is to autonomously path to identified weeds to remove or disrupt them. Initial testing of this robot is completed through a go-to-goal control. This control design takes inputs of the robot position and heading as well as a goal point position. To simulate a GPS system, an OptiTrack motion capture system is used to provide robot heading and position, and machine vision through RGB cameras is used to determine the location of test targets relative to the robot. The controller commands the robot in a way to face towards the goal point, then travel to the goal within a certain radius. This control architecture is shown in the block diagram in Fig 1.

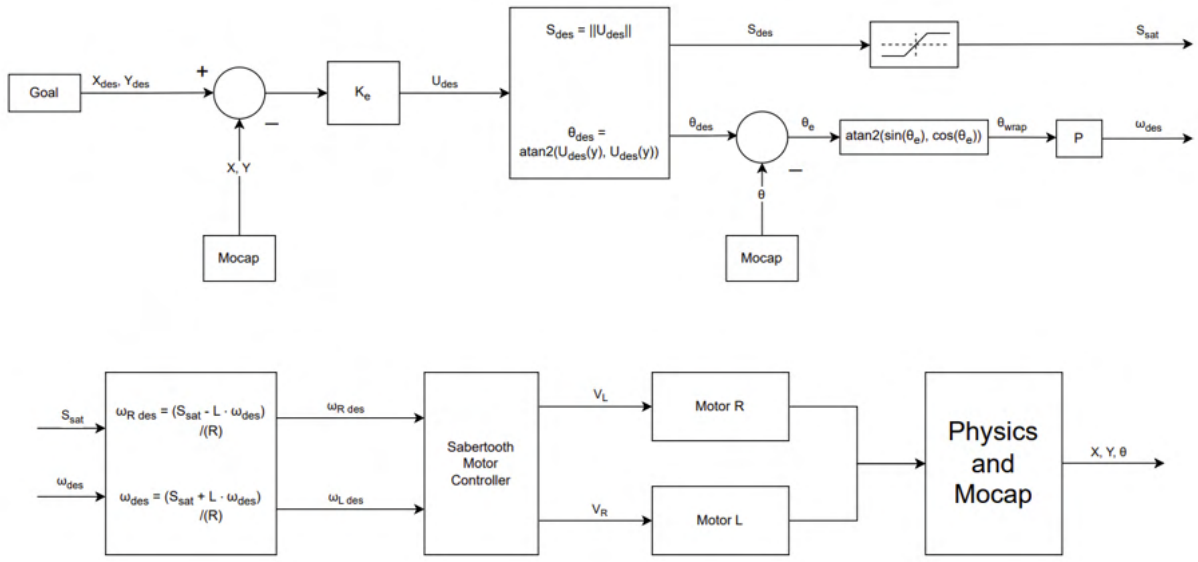


Figure 1. Go-To-Goal System Block Diagram

This control architecture takes the goal location and motion capture information of the robot position, heading, outputs motion, and angle vectors to get the robot to the goal. This is done through several control parameters which are tuned to convert distance and heading vectors into desired directions and speeds, and further into voltage commands for the Sabertooth motor controller.

Additionally, the robot has waypoint path following capabilities. This works similarly to the go-to-goal, as if there were a series of goal points that the robot completes in order. For this functionality, the goal thresholds and other parameters are loosened to prioritize a smooth path over highly precise navigation. This functionality also uses OptiTrack to

get robot position and heading, and utilizes pre-coded waypoint positions for the robot to navigate to, creating a path.

4.2 Mechanical Design

The robot is designed to mechanically remove or disrupt weeds, as opposed to chemical or laser removal. Several removal methods were considered, and two methods were chosen to focus on.

The auger drill bit is designed to remove deep-rooted, matured weeds. This design was tested prior to this year's iteration of the project and was found to be effective at removing weeds with long, deep roots, fully de-rooting the plant. Housing for this auger drill bit was designed to create linear vertical motion. This design uses a stepper motor attached to a lead screw to lift and lower a drill motor which connects the drill bit. This mechanical design is shown below in Fig 2.

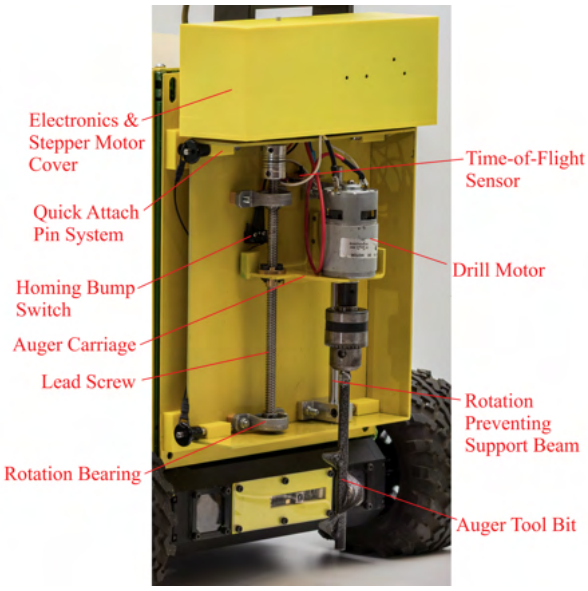


Figure 2. Auger Tool Mechanical Design

After speaking with the project clients and getting an idea of what the potential customers of this product would want, it was decided to also include a tine weeder rake attachment. The rake attachment is a simple pronged design that drags behind the robot during its travel and disrupt smaller, seedling-stage weeds. The attachment mechanism for this tool is identical to that of the auger tool attachment for ease of use. An early prototype of this tool was created in this iteration of the project, and future work on the tool will be beneficial. The prototype of the tine weeder attachment is show in **Figure** 3 below.



Figure 3. Tine Weeder Prototype

4.3 Electrical Design

In order to control the front tool, a new PCB was developed. The layout of this board is shown in Fig 4.

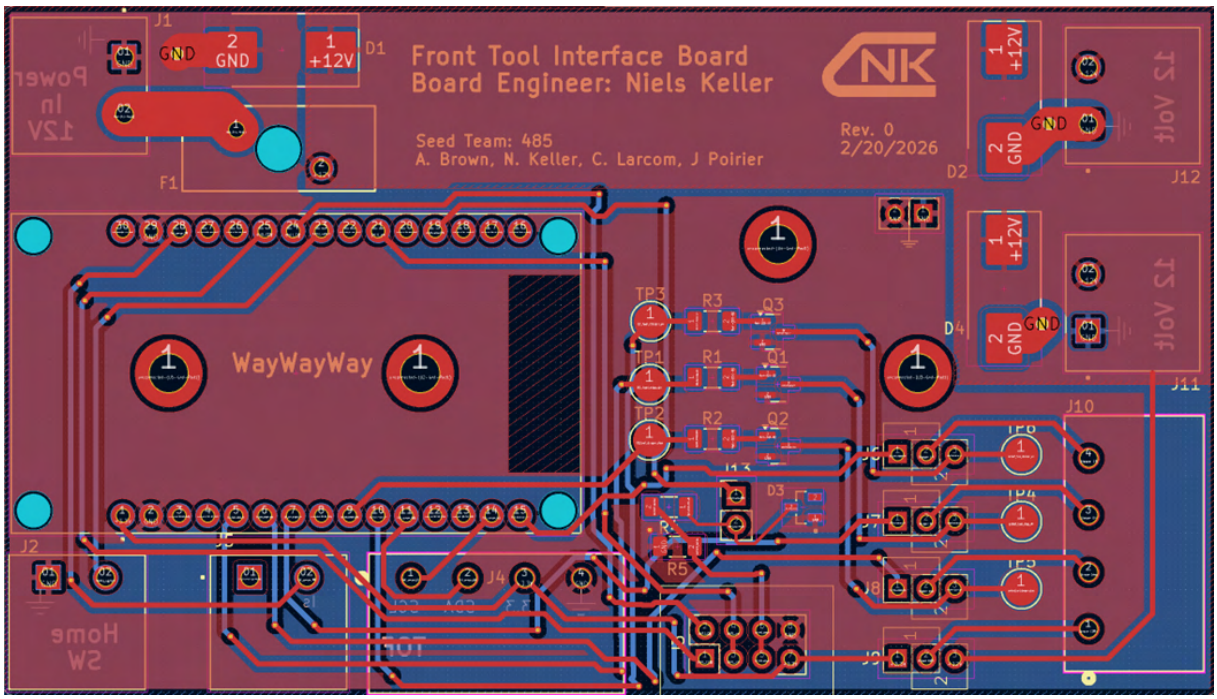


Figure 4. Front Tool PCB

This PCB has 3 major functions:

4.3.1 Power To Components

One of the main functions of this board is to provide power to the stepper driver and the drill. The circuit used for this is shown in Fig 5. It uses the SK810L-TP Shottky diode [1] as a fast acting transient suppressor and reverse polarity protection. This means the diode will short any reversed connection. This can result in very high currents, so a 5 A fuse was inserted to protect any power delivering equipment, as well as the diode itself. In case

of a fault, the diode is rated at 8 A, meaning the fuse will blow significantly prior to this threshold.

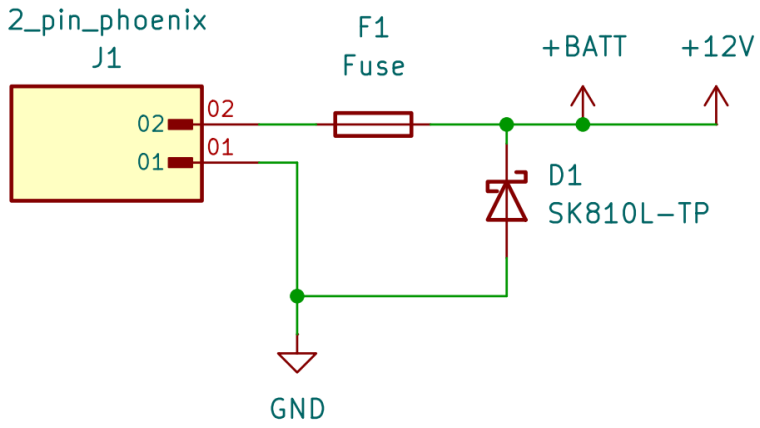


Figure 5. Power input connection schematic

Additionally, these protective diodes were repeated on each power output as shown in Fig 6

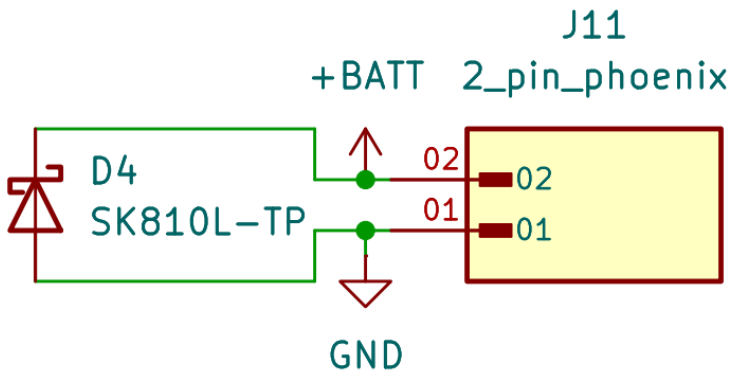


Figure 6. PCB power output schematic

4.3.2 Command Proxy for the Main Computer

The onboard NVIDIA Jetson sends commands to the front board micro-controller via serial. These commands are basic and only describe general actions, allowing the Jetson to dedicate it's computing power to more critical tasks. The front tool board decodes the commands, and handles them. When it is done, it sends a single byte code which indicates the status of the tool.

4.3.3 Digital Communication To Components

In order to execute the commands the front tool receives, it sends digital logic signals to the onboard stepper driver, drill motor driver, distance sensor, and bump homing switch.

The system diagram for the front tool is shown in Fig 7.

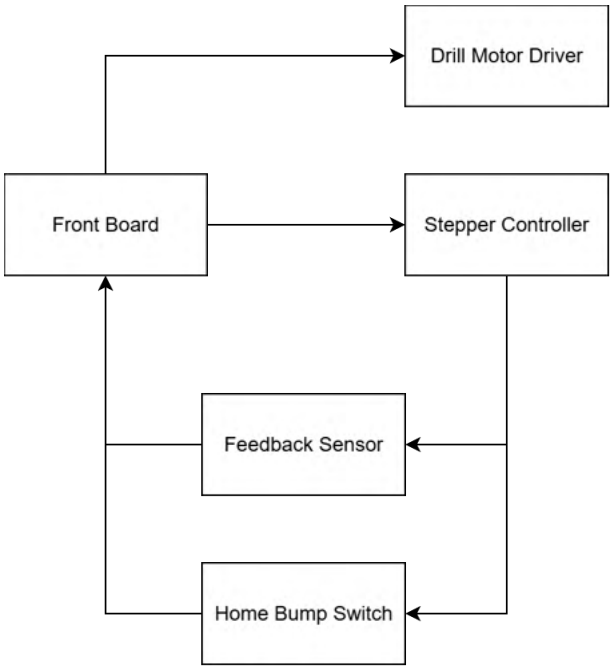


Figure 7. Front Tool System Diagram

4.4 Machine Vision and Plant Identification

In order to fully integrate autonomous control, a machine learning model will be used to autonomously identify weeds in real time. The robot uses an Intel RealSense d435i RGB+Depth camera mounted to the top of the robot to capture images to put through a trained classification model. Additionally, an Arducam 8MP IMX179 camera is used for additional view range and to help cover blind spots of the RealSense. The machine learning model will identify any weeds in the image that it is trained on. Two models have been tested for this iteration of the project. The system overview for the machine vision and machine learning is shown below in Fig 8.

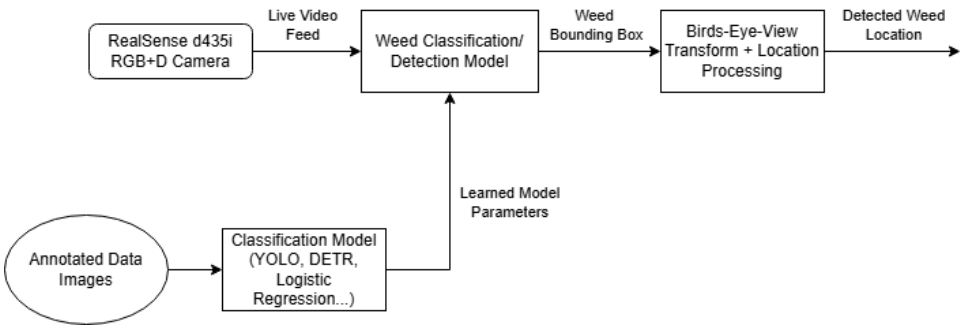


Figure 8. Machine Vision and Machine Learning System Diagram

The first model tested for the implementation of weed classification was a You-Only-Look-Once, or YOLO, model [2]. Testing of this model included training the model on 350 annotated images of dandelions taken with the RealSense camera. These images

were manually collected using the RealSense camera and annotated for dandelions using CVAT [3]. A sample annotated image is shown below in Fig 9



Figure 9. Annotated Image Sample of Dandelions

In addition to the YOLO model, a Detection Transformer model, or DETR model, was also tested [4]. The general structure and training of this model follows the same process as that for YOLO. An analysis on the performance of these models can be found in section 5.1.1.

4.5 Integration

To complete the project, the mechanical, electrical, and software designs must be integrated together to work as one complete unit. A full system summary is shown in the block diagram in Fig 10.

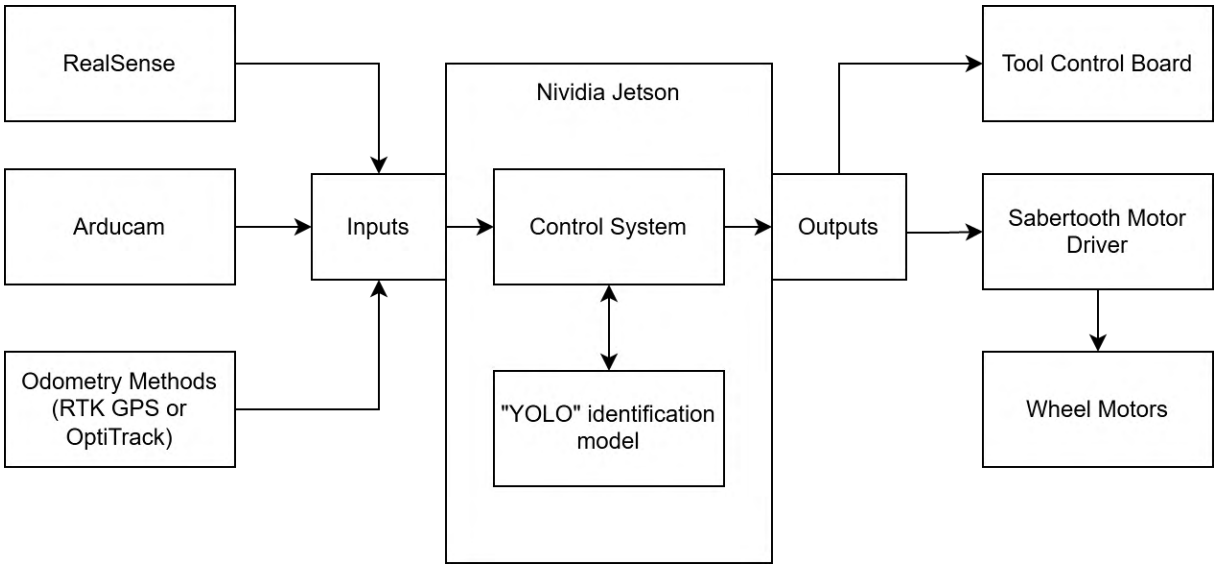


Figure 10. Full System Overview Diagram

The robot takes inputs from a positioning system, in the case of testing this has been an OptiTrack motion tracking system, and the RealSense d435i RGB+D and Arducam RGB cameras. The robot takes pre-defined waypoints to algorithmically create a path to follow, and continuously updates to follow that path. Simultaneously, the live video feed is processed through the machine learning algorithm to detect weeds. If the algorithm detects a weed, the robot will divert from the path following algorithm and move to a go-to-goal algorithm. The go-to-goal will first process the video feed to transform to a birds-eye-view, and then extrapolate the location of the goal (weed) relative to the robot. The algorithm then commands the motors to move towards the goal. The goal location continually updates as the robot moves, until the robot is detected to be at the goal, at which point the end-effector actuation controller will be utilized. Then, once the end-effector actuation has been completed, the robot will revert back to the path following algorithm and return to the path to continue weed detection. The final robot design is shown in **Figure** 11.



Figure 11. Final Auger Weeding Robot

5 Design Evaluation

5.1 Technical Analyses

5.1.1 Machine Learning

In order for the weed classification to work properly, a model must be chosen and trained. There are many model types that exist for image classification, some bare bones algorithms like logistic regression or XGBoost, and some complex model architectures like You-Only-Look-Once (YOLO) and Detection Transformer (DETR). Importantly, the simple algorithms are generally incapable of detecting *where* on the image an object is, only if the image contains the object. Thus, more complex, public model architectures were explored for the purpose of this project.

The most important design decision when it comes to choosing a machine learning model is how well it performs by particular metrics. There are many metrics that can test a model, and for this project, the primary metric used is a confusion matrix. This is a visualization of how many true positives, true negatives, false positives, and false negatives a model predicts given a set of known test data. In this case, a true positive refers to when the model detects an object and the ground truth label agrees. For a true negative, both the model prediction and the ground truth say that there is not an object. For a false negative, the model predicts no object and the ground truth disagrees, and for a false positive, the model predicts there is an object, and the ground truth disagrees. The confusion matrix visually shows how often the model correctly predicts the ground truth. From this confusion matrix, several metrics can also be calculated, as described in the equations below. In these equations, TP is the number of true positives of the model prediction, TN corresponds to true negatives, and FP and FN correspond to false positives and negatives, respectively.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

In this case, accuracy measures the overall correctness of the model, precision measures the purity of positive predictions (i.e. having few false positives), and recall measures the completeness of positive predictions (i.e. having few false negatives). For the context of this project, the most important metric is precision. A false positive in the case of the autonomous weeding robot would mean that the model has detected that there is a weed when there is not actually one, and thus the robot would attempt to actuate on an undesired target. This poses both safety and effectiveness concerns.

The two models that have been analyzed for this project are YOLO and DETR, both of which are complex convolutional neural network architectures. Importantly, although both of these models are capable of predicting the locations of the detected objects within the

images, the testing is being done simply to check if the image has a dandelion or not. This was chosen to simplify preliminary testing in order to get an idea of model performance and because of a relatively limited dataset.

After training both the YOLO and DETR models, they were tested on 145 and 146 images, respectively. None of the testing images were used as training data to ensure that there was limited over-fitting. The reason for the one-image discrepancy between the two models is because DETR utilizes image transforms to create artificial environment condition variance (different saturation, exposure, etc.), which led to an artificial image being implemented in the testing. For each of the models, the confusion matrices are shown in **Figures 12 and 13** and the numerical results are shown in **Table 1**.

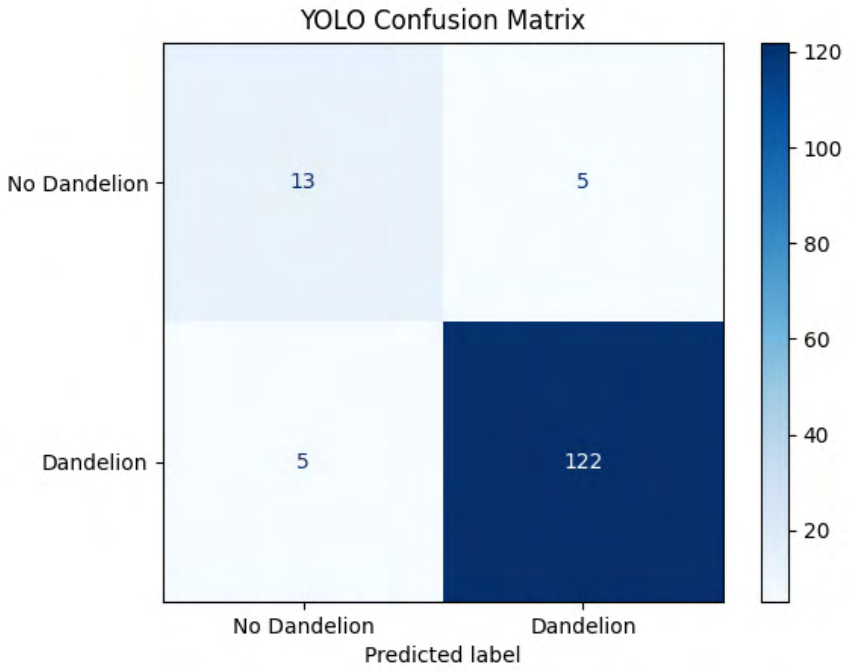


Figure 12. YOLO Confusion Matrix

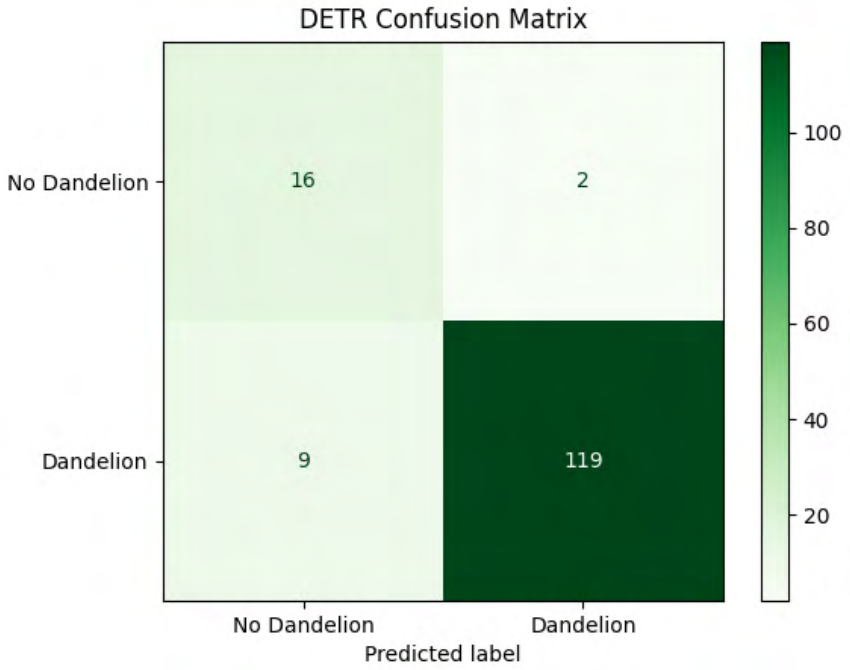


Figure 13. DETR Confusion Matrix

Table 1. Model Performance Metrics

Metric	**YOLO**	**DETR**
Images Tested	145	146
True Positives (TP)	122	119
True Negatives (TN)	13	16
False Positives (FP)	5	2
False Negatives (FN)	5	9
Accuracy	93.10%	92.47%
Precision	96.06%	98.35%
Recall	96.06%	92.97%

For this project, of course, a high number of true positives and true negatives is ideal, but the most important metric is the false positives, as previously explained. Having that in mind, both YOLO and DETR perform well in regards to getting true classifications, showing effectiveness from both. However, DETR had a lower number of false positives, despite a slightly lower overall performance. Since safety is a forefront concern, the slight dip in overall effectiveness can be deemed worth the increased safety factor of less false positives, and thus it can be concluded that under these test conditions, DETR is the stronger choice of model for this project.

These promising results allow the project to move forward with implementing the model to classify *where* the dandelions are within the image, as opposed to just if they are present in the image. This will be crucial for the future of this project, as the relative position of the detected weed must be calculable in order to control the robot to move towards the weed.

For this stage of the project, DETR proved to have complications interfacing with the hardware. Due to time constraints, YOLO was used for its ease of implementation, but DETR does have slightly stronger promise. More models, including more modern or advanced models, can and should be explored in the future.

5.1.2 Motor Sizing

An auger bit will be used to drill out the weeds, the current system that is implemented is using a recycled drill motor. The goal of this analysis is to confirm that the motor that is currently implemented will be able to meet the project's power requirements for the application. There are several important variables that affect this outcome: auger lead angle, axial force on ground, and soil density. These factors are then used to find the torque needed for the motor which is the primary metric being used for selection, as most electric motors have abundant speed for this operation. To find this torque, a simplified power screw approach combined with soil cutting mechanics will be used. In this analytical model, the cutting tool must overcome the friction from the soil as well as the soil cohesion.

1. c = Soil Cohesion = 20 kPa
2. d = Diameter = .0508 m
3. Pitch = .025 °
4. α = Lead Angle = 18 °
5. μ = Soil Friction Coefficient = .45
6. γ= Soil Density =1800 kg/m^3
7. $F_{axial} = 150N$

The swept contact area first needs to be calculated using Equation 4. Using the constants above, the swept area is found to be 0.004 m^2. The cutting force is then calculated with Equation 5, resulting in 83.6 N. Finally, the cutting force value is used in Equation 6 to find that the motor powering the auger needs to produce 10.76 N×m.

$$A_{blade} = \pi \times d \times \text{pitch} \tag{4}$$

$$F_{cut} = c \times A_{blade} + \gamma \times d \times A_{blade} \cdot \tan(\phi) \tag{5}$$

$$T = (F_{cut} + F_{axial}) \times \frac{d}{2} \times \frac{\tan(\alpha) + \mu}{1 - \mu \times \tan(\alpha)} \tag{6}$$

The DeWalt drill motor has a maximum torque of 42 N×m, which provides a comfortable margin of 3.9× above the calculated requirement of 10.76 N×m. One limitation of this calculation is that it does not account for the additional resistance encountered when cutting through weed root systems. This analysis provided motivation to explore this motor experimentally. When tested on developed weeds, it was able to pass all tests that were applied which made it a suitable choice for further use in the project.

5.1.3 Stress Calculation

A lead screw will be used to provide precise, smooth control of the auger. This mechanism allows a relatively small stepper motor to generate the torque required for soil penetration while maintaining stable, reliable motion.

Screw loading calculations were performed to validate the selection of the lead screw for the robot's auger. The auger is made of mild steel which has a yield strength of roughly 250 - 350 MPa (the median of 300 MPa was used for the calculations). Screw loading refers to the forces acting on a screw during operation. As axial load increases, frictional and mechanical resistance within the drive system also rise. Consequently, the stepper motor must supply additional torque to hold its position and to drive the auger with the required motion.

Shear stress was calculated using a range of axial thrusts (5 N to 100 N) to represent the variety of soil forces the auger may encounter. The lead angle was determined from the selected lead screw, and the equations below were used to compute axial stress, thread shear stress, torsional shear stress, and thread bearing stress.

$$\lambda = \arctan\left(\frac{L}{\pi \cdot d_m}\right) \tag{7}$$

$$T = \frac{F \cdot d_m}{2} \cdot \frac{\tan(\lambda) + \mu}{1 - \mu \cdot \tan(\lambda)} \tag{8}$$

$$\sigma_{axial} = \frac{4F}{\pi \cdot d_r^2} \tag{9}$$

$$\tau_{torsion} = \frac{16T}{\pi \cdot d_r^3} \tag{10}$$

$$\tau_{shear} = \frac{F}{\pi \cdot d_r \cdot p \cdot n} \tag{11}$$

$$\sigma_{bearing} = \frac{2F}{\pi \cdot d_m \cdot p \cdot n} \tag{12}$$

$$\sigma_{vm} = \sqrt{\sigma_{axial}^2 + 3\tau_{torsion}^2} \tag{13}$$

$$\sigma_{vm} < \frac{\sigma_y}{SF} \tag{14}$$

The results indicate that the maximum stress on the lead screw is under 3 MPa, which is far below the material's approximate 300 MPa stress capacity. Under the expected full load of 15 N, the resulting stress is nearly negligible relative to the screw's maximum capability. Even when assuming a harsh operating environment and applying a safety factor of 4 (corresponding to an axial thrust of 60 N), the lead screw remains more than capable of withstanding the predicted stresses. **Table 3** summarizes the values used in these calculations. Most parameters were taken directly from the specifications of the selected lead screw, while the friction coefficient and safety factor were chosen based on the screw's material and the expected operating conditions.

Table 2. Stress Calculations for Lead Screw Loading

Axial Thrust (F)	Lead Angle	Torque Required (Nm)	Axial Stress (MPa)	Shear Stress (MPa)	Thread Bearing Stress (MPa)	Von Mises Combined Stress
5	0.334	0.011	0.142	0.183	0.018	0.347
10	0.334	0.022	0.284	0.365	0.036	0.694
15	0.334	0.032	0.425	0.548	0.054	1.040
20	0.334	0.043	0.567	0.731	0.072	1.387
25	0.334	0.054	0.709	0.913	0.090	1.734
30	0.334	0.065	0.851	1.096	0.108	2.081
35	0.334	0.076	0.993	1.279	0.126	2.427
40	0.334	0.086	1.135	1.462	0.144	2.774
45	0.334	0.097	1.276	1.644	0.162	3.121
50	0.334	0.108	1.418	1.827	0.180	3.468
55	0.334	0.119	1.560	2.010	0.198	3.814
60	0.334	0.129	1.702	2.192	0.217	4.161
65	0.334	0.140	1.844	2.375	0.235	4.508
70	0.334	0.151	1.985	2.558	0.253	4.855
75	0.334	0.162	2.127	2.740	0.271	5.201
80	0.334	0.173	2.269	2.923	0.289	5.548
85	0.334	0.183	2.411	3.106	0.307	5.895
90	0.334	0.194	2.553	3.288	0.325	6.242
95	0.334	0.205	2.695	3.471	0.343	6.588
100	0.334	0.216	2.836	3.654	0.361	6.935

Table 3. Lead Screw Geometric and Material Parameters

Parameter	Value
Pitch (m)	0.002
Lead (m)	0.008
Mean diameter (m)	0.00735
Root diameter (m)	0.0067
Friction coefficient	0.20
Number of engaged threads	4
Yield strength (MPa)	300
Safety factor	4

5.1.4 Odometry Method

In order for the control systems to work, the controller requires positional feedback. Also known as "Odometry", this allows the controller to calculate what direction it should be going. There are multiple ways to obtain this data, each with different pros and cons. As such a Pugh matrix was used. There are four main considerations:

1. **Cost**: Cost is a factor in both this year's budget, as well as a unit cost of any

finalized product. One of the overarching goals of this project is to provide a low cost alternative to existing products.

2. **Accuracy**: An important factor which determines the quality of the way-point tracing system is the accuracy of the odometry. Quite simply, if inaccurate data is fed to the controller, it misjudges how far, and at what angle the goal is with respect to the heading of the robot.
3. **Ease of Use**: Because the scope of this year's project is mainly to develop the autonomy, it is vital that the positioning system is reliable and easy to setup.
4. **Ability to use on farm**: One large aspect to consider is how easy it will be to implement this system on an actual garden or farm.

The two alternatives considered were RTK GPS and OptiTrack [5] (an optical tracking system). A pro/con list for both options is shown in **Table 4**. In order to balance these considerations a Pugh matrix was used. Each aspect was graded one a scale of 1-10. This is shown in **Table 5**. Overall, the OptiTrack system beats RTK GPS primarily due to the ease of use and cost. While we will be unable to use this system in the final product, the scope of this year's work is primarily focused on getting the system working in a laboratory environment. This is to make sure that the control system can be properly implemented. Since RTK GPS will be required for outdoor implementation in the future, it has been ordered with the remaining budget for this year to give the next iteration of the project a good starting point.

RTK Pro	RTK Con	OptiTrack Pro	OptiTrack Con
High accuracy	High cost	High accuracy	Only works in lab
Designed for Outdoors	Doesn't work in lab	Infrastructure exists in lab	Difficult to initially implement
Highly relevant to end goal	Very difficult to implement	No additional cost	

Table 4. Pro/Con List

Category	**RTK GPS**	**OptiTrack**
Cost	1	10
Accuracy	10	10
Ease Of Use	1	7
Ability to use on farm	10	1
Total	22	28

Table 5. Predicted Future Expenses

6 Engineering Specifications Evaluation

This project passed each of the design specifications. The pass rate is 100%. Each of the specifications are shown in Table 6.

Specification	Status
(10) Waypoint Following	Pass
(20) Camera Go-To-Goal	Pass
(30) Tool Soil Test	Pass
(40) Dandelion Detection (Accuracy)	Pass (93.10%)
(50) 2 Tools Made	Pass
(60) Cost < $5,000	Pass ($2,309.10)
(70) Remote Stop	Pass
(80) Front Tool Speed	Pass (40 seconds)
(90) Dandelion Detection (Precision)	Pass (96.10%)
(100) Optical Test Target Detection	Pass

Table 6. Engineering Specifications

7 Budget Analysis

Provided by clients, Dr. Hamid Ossareh and Dr. Luis Duffaut, the budget for this project was $1,000.00. During this year (2025-2026), 72.72% ($727.24) of the budget was used, leaving a comfortable buffer in the end. Spending felt comfortable among the team where the budget did not feel restraining while allowing the team to make quality purchases. Any necessary purchases were made as soon as the necessary product was identified, to avoid as much delay as possible. The SEED lab was checked for any parts that could be found to save money. A recommendation for future teams would be to distribute expenditures more evenly throughout the academic year. Although fiscal caution is a commendable trait, the effort devoted to minimizing costs may be better directed toward other aspects of the project. Since a quarter of the budget was still unused, the current team decided to make a future purchase of an RTK GPS for next year's SEED Team to set a strong foundation in the beginning of the year.

Project Budget

Date	Categories	Item	Unit Cost	Units	Total Cost	% of Total Budget
09/27/25	Planter	BOOST LIGHTING 22.0-in 1-Light Black 16-Watt Full Spectrum LED Grow Light Kit	$21.83	2	$43.66	4.37%

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Date	Categories	Item	Unit Cost	Units	Total Cost	% of Total Budget
09/27/25	Planter	Miracle-Gro Organic 1.5 Cubic Feet All-Purpose Organic Raised Bed Soil	$9.48	2	$18.96	1.90%
09/27/25	Planter	Utilitech 15-Amps 125-Volt Indoor or Outdoor 1-Outlet Countdown Analog Lighting Timer	$6.63	1	$6.63	0.66%
09/27/25	Planter	Project Source 10-Feet x 20-Feet Clear Light-Duty Plastic Sheeting	$4.73	1	$4.73	0.47%
09/24/25	Planter	Seed Needs Dandelion Seeds for Planting Taraxacum officinale — Non-GMO Heirloom & Untreated Garden Seed to Plant Dandelions for Herbal Uses & Edible Salads (2 Packs)	$3.49	1	$3.49	0.35%
10/06/25	Electrical	MCA20B-V4E30R Emergency Stop SW, 3PST-NO, 120V, Screw	$4.04	1	$4.04	0.40%
10/26/25	Mechanical	Nut from Fastenal	$0.16	1	$0.16	0.02%
11/07/25	Mechanical	Lead Screw & Support Kit	$29.99	1	$29.99	3.00%
12/04/25	Lab	Safer Home Indoor Plug-In Fly Trap SH502 — Effective Fruit Fly Traps for Indoors, Gnat Traps for House Indoor, 400 Sq Ft Protection — 1 Device + 2 Glue Cards	$8.66	1	$8.66	0.87%
12/04/25	Lab	Safer Home SH512 Trap Refill Pack of Glue Cards for SH502 Indoor Fly Trap, 12 Count (Pack of 1)	$16.41	1	$16.41	1.64%

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Date	Categories	Item	Unit Cost	Units	Total Cost	% of Total Budget
01/23/26	Electrical	3 PCS TB6600 Stepper Motor Driver, ABuff 5A 9–40V Nema 17 Stepper Motor Driver CNC Controller Single Axes Hybrid Stepper Motor Controls	$24.99	1	$24.99	2.50%
01/23/26	Electrical	Adafruit VL6180X Time of Flight Distance Ranging Sensor (VL6180) — STEMMA QT	$13.95	1	$13.95	1.40%
01/26/26	Mechanical	2 Pack 304 Stainless Steel Quick Release Pin 0.24 Inch Diameter Ball Locking Pin with Push Button Round Handle Lanyard, Usable Length 1.18 in (6mm x 30mm)	$14.99	2	$29.98	3.00%
01/26/26	Electrical	ESP-WROOM-32 ESP32 Development Board 2.4GHz Dual-Mode WiFi + Bluetooth Dual Cores Microcontroller, Integrated Antenna RF AMP Filter AP STA, Arduino IDE Compatible (3PCS)	$17.99	1	$17.99	1.80%
02/02/26	Housing	0.09 in Aluminum Sheet 5052-H32 — Part #: 7130	$18.85	5	$94.25	9.43%
02/09/26	Electrical	10 Pcs Micro Limit Switch for 3D Printer Endstop, Precision Mini Limit Switches Replacement for Print Bed Leveling and Control Board	$12.31	1	$12.31	1.23%
02/23/26	Electrical	PCB Front	$27.22	1	$27.22	2.72%
02/23/26	Electrical	Digi-Key PCB 1 Components	$134.70	1	$134.70	—
03/10/26	Housing	0.09 in Aluminum Sheet 5052-H32 (12 in x 12 in)	$15.65	2	$31.30	3.13%

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Date	Categories	Item	Unit Cost	Units	Total Cost	% of Total Budget
03/16/26	Electrical	Arducam 8MP Autofocus USB Camera, IMX179 Lightburn Camera with Microphone, Ultra HD Embedded Webcam, Compatible with Linux, Windows, and Mac OS	$33.99	1	$33.99	3.40%
03/20/26	Electrical	Digi-Key PCB 2 Components	$39.10	1	$39.10	3.91%
03/22/26	Housing	7 Rolls of High Visibility Tape	$9.99	1	$9.99	1.00%
03/23/26	Housing	6 ft T-Slotted Framing Aluminum Extrusion Single Four Slot Rail, Silver, 20mm Square, Solid	$24.69	1	$24.69	2.47%
03/23/26	Housing	20 Sets Black 2020 Series Aluminum Extrusion Corner Bracket Kit, Includes 20 pcs Right Angle Connectors, 40 pcs M5 T-Slot	$14.99	1	$14.99	1.50%
03/27/26	Housing	Krylon Green Spray Paint	$6.63	1	$6.63	0.66%
03/27/26	Housing	Krylon Yellow Spray Paint	$6.63	1	$6.63	0.66%
03/27/26	Mechanical	2-1/2 in Stainless Steel Nails	$16.12	1	$16.12	1.61%
03/27/26	Housing	Sealant	$8.44	1	$8.44	0.84%
03/28/26	Mechanical	2 Pack 304 Stainless Steel Quick Release Pin 0.24 Inch Diameter Ball Locking Pin with Push Button Round Handle Lanyard, Usable Length 1.18 in (6mm x 30mm)	$14.99	2	$29.98	3.00%
04/03/26	Housing	Krylon Green Spray Paint	$6.63	1	$6.63	0.66%
04/03/26	Housing	Krylon Yellow Spray Paint	$6.63	1	$6.63	0.66%
04/25/26	Electrical	RTK GPS	$190.00	1	$190.00	19.00%

8 Retrospective

Over the course of the past two semesters, the team has learned a lot about team management. One of the team's most important process changes was more strictly enforcing staying on topic during scrums. This was important because it greatly improved how efficient scrums were.

Before implementing this change, team scrums were typically much longer, and did not convey all the information they should have. After the change, not only was all important information discussed, but it also happened much quicker allowing the team to break into smaller groups and get to work much faster.

After this change, this was one aspect of our scrum that worked very well. One part that the group thought didn't work as well was the frequency of scrum huddles. Overall it could be improved by increased one on one communication and then reducing the number of dedicated scrum huddles.

9 Review/Recommendations

While this year has been recognized as exceeding the expectations of the clients and instilling great satisfaction in the team, the project is still not finished. This is as expected, and there is already a SEED project planned to continue this work. The following are some of the areas that need improvement, in order of priority.

9.1 Implement RTK GPS

RTK GPS is the primary required next step in order to further this project. This year's work has setup much of the foundation required to make RTK GPS work. This was done by providing a software pipeline for odometry.

9.2 Improve Software Integration

In its current state, the robot has several successful modes of operation in a lab setting. However, the software is not currently designed for the variance that comes with outdoor usage. Additionally, the modes of operation are currently segmented, and have limited inter-operation capabilities. Further work on creating a full software pipeline will be crucial.

9.3 Improve/Expand Tools

The auger tool had monumental improvements made to it from the previous iteration of this project. However, there is still room to grow, especially for additional tools. The auger drill bit that was developed in the previous iteration and used throughout this iteration is unbalanced and could use updating. A prototype of a tine weeder attachment was created in this iteration of the project, and further work on this tool would be beneficial. Additionally, the team recommends the development of a finger weeder attachment for future iterations of the project.

9.4 Modify the Frame Internals

Currently it is difficult to access the battery and electronics. Adding a quick release to the back/top panels would help remedy this. Furthermore, the internal electronics holders could be reprinted to run the cables in a more organized manor.

9.5 Change Drill Motor

The drill motor used for the auger tool is currently over-powered and unbalanced. The shaft of the drill is slightly bent, causing significant vibrations during operation. The current drill motor also spins much faster than required, even running at as low as 5% power, and could be specified better for the use of this robot.

10 Lesson Learned

A major lesson the team learned through this project is to keep documentation updated as you go. As the project came to a close, the team had to rush to get full assembly drawings, file organization, code script descriptions, and electrical drawings to be a suitable for a finalized project. Though the team always intended to do this and understood the importance of it, especially considering this project will continue, the effort and time put into documentation was severely underestimated. This impacted the project by limiting the amount of last-minute improvements that could be made, since more time was focused on documentation. This also affected the project schedule in the end of the project life cycle. For future engineering students, the team recommends keeping documentation of every aspect of the project throughout, even if it is not in an official, presentable format.

References

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